



Sai Krishna Sangeetha

Senior Full-Stack Developer

saikrishna.sangeetha@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | (689) 250-9414 | Tampa Bay, FL

PROFESSIONAL SUMMARY

- Senior Full-Stack Engineer with **6+ years of experience building large-scale SaaS platforms** using **React, Python (Django/FastAPI), and AWS**. Specialized in designing **high-throughput distributed systems, real-time data platforms, and AI-driven observability solutions**.
- Expert in **React architecture (TypeScript, Zustand/Redux, Tailwind)** for building scalable frontends handling complex data flows and real-time dashboards. Proven ability to optimize rendering performance, state management, and component architecture for **high-performance UI systems**.
- Strong backend engineering expertise using **Python, Django, FastAPI, Redis, Celery**, and event-driven architectures, building high-throughput APIs, asynchronous processing pipelines, and distributed workflows.
- Experienced in **ML-driven systems and anomaly detection pipelines**, implementing Isolation Forest models, semantic analysis, and AI-assisted remediation pipelines for large-scale log data.
- Architect of **cloud-native systems on AWS**, implementing containerized services, serverless APIs, scalable storage pipelines, and automated CI/CD using Docker, Kubernetes, and GitHub Actions.
- Deep expertise in **observability platforms and telemetry pipelines** using Elasticsearch, Kafka, Fluent Bit, Open Telemetry, and Prometheus, processing millions of events daily for monitoring, anomaly detection, and root-cause analysis.
- Strong focus on **system reliability, security (OAuth2, SAML, JWT, RBAC), and database performance tuning** across PostgreSQL, and MySQL.

CORE TECHNICAL SKILLS

Frontend Engineering: React.js, TypeScript, JavaScript (ES6+), Redux, Zustand, Tailwind CSS, Webpack, React Router, Component Design Systems, React Flow

Backend Engineering: Python, Django, FastAPI, Golang, REST APIs, GraphQL, Microservices, Async Programming, Redis, Celery

AI/Data Systems: Anomaly Detection Pipelines, Isolation Forest, Semantic Autoencoders, LLM-based automation, Data Processing Pipelines, Feature Engineering

Data & Databases: PostgreSQL, MySQL, Elasticsearch, SQL Alchemy, Django ORM, Query Optimization, Indexing Strategies

Distributed Systems & Messaging: Kafka, Temporal, Event-Driven Architecture, Streaming Pipelines, Batch Processing

Observability: Open Telemetry, Elasticsearch, Kibana, Logstash, Fluent Bit, Prometheus, APM Systems, Alerting Pipelines

Cloud & Infrastructure: AWS (Lambda, API Gateway, EC2, S3, CloudFront, Cognito), Docker, Kubernetes, Pulumi, IaC

CI/CD & DevOps: GitHub Actions, Jenkins, GitLab CI, CircleCI, CodePipeline, Automated Deployments

Security & Authentication: OAuth 2.0, JWT, SAML, SSO, RBAC, Multi-Tenant Access Control

Methodologies & Tools: Agile (Scrum/Kanban), Git, Jira, Figma, Postman, Unit Testing, Design Patterns, TDD/BDD, SAAS Application Development, Performance Optimization, SEO Best Practices, Responsive Web Design, Component Libraries

EXPERIENCE

- Senior Software Developer | Ceburu Systems Inc., – Pleasanton, CA (Remote)** **Oct 2024 – Present**
 - Architected and built a **distributed observability platform** supporting logs, metrics, traces, and network telemetry using **Open Telemetry, Kafka, Logstash, and Elasticsearch**, enabling large-scale monitoring and root-cause analysis across distributed environments.
 - Designed **edge telemetry agents in Go** for Linux and Windows to securely collect system logs, metrics, NetFlow, and application traces, streaming telemetry to cloud pipelines with **secure enrolment, version control, and health monitoring**.
 - Built high-throughput streaming pipelines ingesting **millions of telemetry events per day**, implementing fault-tolerant processing, replay capability, backpressure control, and scalable alerting workflows.
 - Developed **AI-driven anomaly detection pipelines** for large-scale log data stored in S3, orchestrating **Isolation Forest, semantic autoencoder models, and rule-based detectors** with batch processing and automated remediation generation.
 - Implemented **dynamic service topology visualization** using React + React Flow, enabling real-time service dependency mapping and performance insights.
 - Built **high-performance analytics APIs using Django and Elasticsearch DSL**, enabling KQL-style log search, real-time alerting, and distributed query execution across large telemetry datasets.
 - Designed and implemented **multi-tenant RBAC architecture** supporting hierarchical module visibility and granular user access controls.
 - Developed an **enterprise IPAM microservice** for IPv4/IPv6 allocation, subnet discovery, and reconciliation, improving IP utilization across enterprise networks.

- Implemented **ServiceNow integration pipelines using REST APIs and Webhooks**, synchronizing incidents and reducing manual operational effort by 40%.
- Led a team of **5 engineers**, defining architecture decisions, reviewing code, and delivering complex features across multiple product releases.

○ **Senior Software Developer** | *AgWise LLC., – St. Petersburg, FL*

Mar 2024 – Sep 2024

- Led full-stack development of a **cloud-native AgTech platform** using **React, Python (Django), and MySQL**, enabling real-time agricultural analytics and nutrient recommendation workflows for multi-user environments.
- Architected scalable **data ingestion and processing pipelines** for soil samples, plant tissue tests, and agronomic inputs, enabling automated nutrient recommendations and real-time decision support for growers.
- Designed and implemented **serverless ETL pipelines** using **AWS Lambda and Step Functions** to automate ingestion and transformation of lab test data from email-based submissions, improving operational efficiency, increasing platform revenue by **30%**.
- Migrated legacy REST endpoints to **GraphQL APIs**, enabling flexible client-side data querying and reducing average response latency by **up to 85%**.
- Built **high-performance, responsive React interfaces** using **React, Redux, React Router, and SASS**, supporting real-time visualization of agricultural analytics and improving frontend maintainability and performance.
- Integrated third-party services including **Stripe for payment processing** and **Mapbox for geospatial visualization**, enabling location-aware analytics and subscription-based product offerings.
- Implemented **automated CI/CD pipelines using AWS CodePipeline**, containerized services using Docker, and orchestrated deployments using **Kubernetes with auto-scaling and load balancing**, ensuring reliable, zero-downtime production releases.
- Optimized application performance through **query tuning, caching strategies, and efficient state management**, reducing application load times by **25%**.

○ **Graduate Research Assistant** | *CECS Department, UCF – Orlando, FL*

Jan 2024 – May 2024

- **Developed responsive, real-time React-based interfaces** with dynamic search and interaction features; optimized data fetching via GraphQL to reduce API overhead, improving application performance, scalability, and efficiency.
- **Conducted research on Java Modelling Language (JML)** under Prof., Gary T. Leavens, exploring formal specification techniques, behavioural interface specifications, runtime assertion checking for Java programs to enhance software reliability, verifiability.

○ **Software Development Engineer** | *Freshworks Inc., – Chennai, India*

Jan 2019 – July 2022

- Engineered scalable backend services for a **high-traffic multi-tenant HR SaaS platform** using **Python, Django, REST APIs, and Redis caching**, significantly improving request throughput and reducing database load by **30%**.
- Designed and deployed **45+ serverless APIs** using **AWS Lambda and API Gateway**, enabling elastic scaling and high availability for global SaaS workloads.
- Architected **event-driven microservices workflows using Apache Kafka**, enabling asynchronous processing for HR events such as candidate updates, onboarding workflows, and notifications.
- Implemented **fault-tolerant asynchronous background processing pipelines** using Redis-backed job queues to handle bulk operations such as candidate imports, email notifications, and reporting tasks at scale.
- Refactored monolithic services into **modular microservices**, improving system maintainability, deployment flexibility, and operational observability.
- Designed cloud infrastructure using **AWS EC2, S3, and CloudFront**, enabling the platform to seamlessly handle **10× traffic spikes** during peak usage periods.
- Implemented **frontend performance optimizations** including lazy loading, client-side caching, and background processing, improving application responsiveness by **40%**.
- Designed and implemented **data-intensive dashboard interfaces for HR workflow analytics**, optimizing client-side data processing, incremental rendering, and asynchronous API interactions to efficiently handle large datasets and reduce UI latency.

SYSTEM DESIGN & ARCHITECTURE

- Distributed log processing pipelines
- Event-driven microservices architecture
- AI-driven anomaly detection systems
- Multi-tenant SaaS platform architecture
- Observability platforms and telemetry pipelines

EDUCATION

- **University of Central Florida** – *Masters in Computer Science*
- **SASTRA University** – *Bachelors in Information Technology*

May 2024

May 2019

PROJECTS

○ GPT-Powered Chatbot

- Developed a **full-stack AI chatbot platform** with a React-based conversational interface and a FastAPI backend handling prompt orchestration and response streaming.
- Integrated **OpenAI GPT models** to generate context-aware responses, implementing conversation history management and prompt structuring for improved response relevance.
- Implemented **Redis-based caching and session management** to reduce response latency and support concurrent user interactions.
- Designed scalable backend services with **PostgreSQL persistence and asynchronous request handling** to support high-volume chat interactions.

○ Online Workers Hiring System

- Designed and implemented a **web-based workforce management platform** supporting user onboarding, job postings, task delegation, and workflow tracking.
- Built modular frontend components and dynamic forms to support **real-time job assignment and worker task management workflows**.
- Implemented secure **authentication and role-based user management**, enabling administrators to manage workforce assignments and operational workflows efficiently.

○ Deep Neural Network Compression

- Developed a **model compression framework for convolutional neural networks**, introducing a pruning strategy guided by explainable AI metrics and transfer learning techniques.
- Reduced model size by **90% while maintaining 83% accuracy**, enabling deployment of deep learning models on **resource-constrained edge devices**.
- Evaluated pruning strategies across multiple CNN architectures to identify optimal trade-offs between **model performance and computational efficiency**.

○ Abusive Comments Identification

- Built a **multilingual abusive content detection model** using BERT architecture trained on the MURIL dataset for Indian languages.
- Designed preprocessing pipelines for multilingual text normalization and tokenization to improve model accuracy.
- Achieved **81% classification accuracy**, enabling automated moderation of abusive content across multilingual social media datasets.

○ Cache Simulator

- Developed **configurable cache simulator** to analyse L1 cache performance across varying sizes (1KB–1MB) and associativity levels.
- Implemented multiple cache replacement policies and analysed their impact on **cache miss rate and average access time (AAT)**.
- Identified **Optimal Replacement policy** as the most efficient configuration for a 32KB cache, reducing miss rate by **45%** and improving access time by **30%**.

PUBLICATIONS

- **Various Dynamic Programming Methods to Solve Optimization Problems:** This paper explores multiple **optimization problems** and their algorithmic solutions. It addresses **the Longest Common Subsequence problem** and **Floyd-Warshall algorithm** while highlighting dynamic programming's essential elements: **optimal substructure** and **overlapping subproblems**.

CERTIFICATIONS

- [2024] **Angular** certified from Hackerrank.
- [2023] AWS Certified **Solutions Architect-Associate**.
- [2023] **Responsive Web Design** certified from freecodecamp.org.
- [2021] **Agile Project Management; Agile Project Delivery** certified from Udemy.
- [2020] **Java(Intermediate)** certified from Hackerrank.

AWARDS & ACHIEVEMENTS

- Awarded “**Will it Make the Boat Go Faster**” at Freshworks Inc., for delivering impactful initiatives and team collaboration.
- Awarded “**Annual Sprint Champion-2020**” at Freshworks Inc., for outstanding contribution.
- Secured a place in the top **5** in the **Hackathon** at Freshworks Inc., showcasing strong problem-solving, collaboration skills.
- Secured **first place** in the **Business Idea Presentation** event of **E-Cell** at SASTRA University.